Project 1: Data Mining Pipeline Technical Report

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Course: Team Project - Semester 4

Section 1: Introduction

A data mining pipeline is a structured, end-to-end framework that converts raw, unorganized data into clear, systematic intelligence. Real-world datasets are usually messy, incomplete, and noisy. Therefore, a systematic series of steps is required to ensure data quality and extract valuable patterns for business or research goals.

This technical report documents our Semester 4 Team Project, which focuses on designing and deploying scalable data pipelines across five different domains. Each data pipeline follows the standard life cycle: objective understanding, source text mining or API collection, structured preprocessing, feature transformation, pattern mining, and visual reporting.

Section 2: Task 1: Web Scraping of Job Listings

A. Objective and Data Mining Target

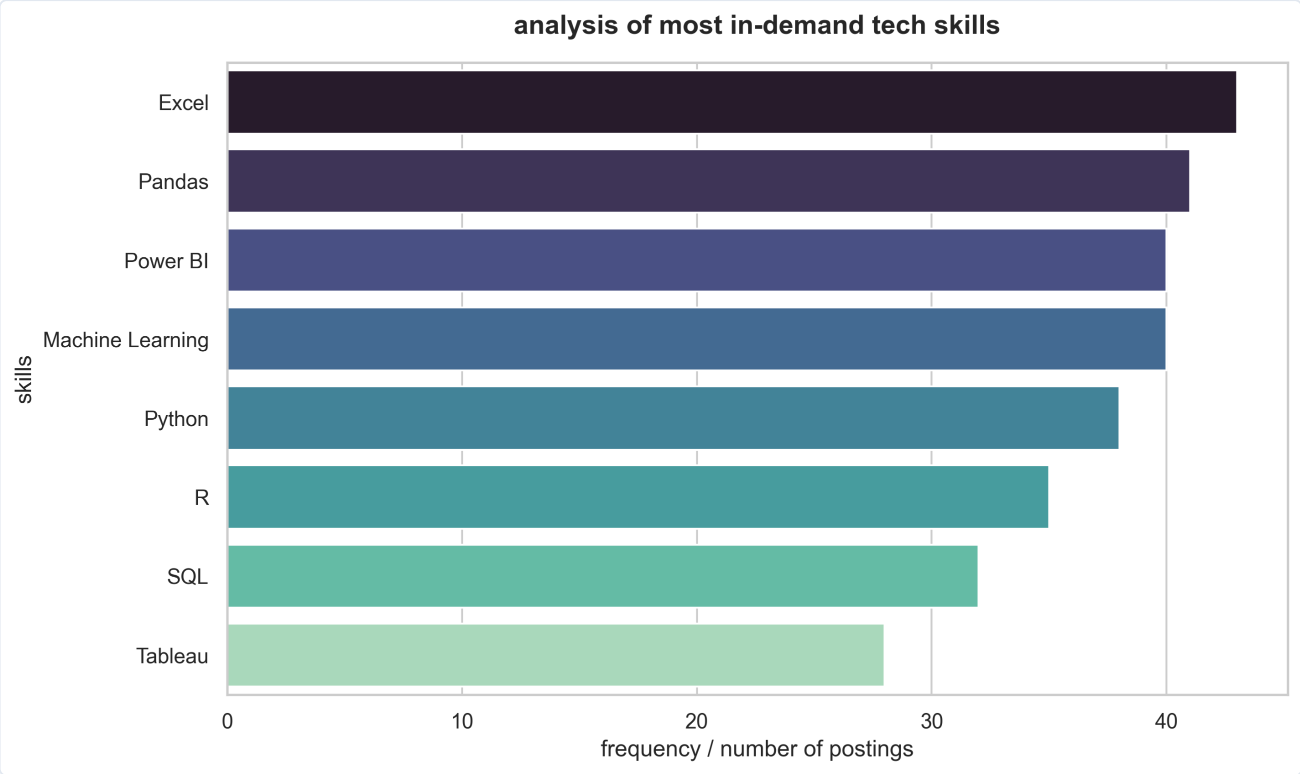
The objective of this sub-pipeline is to evaluate the technical skills that are currently in high demand across the tech job market. By monitoring modern job advertisements, the system attempts to capture hiring patterns and determine which technologies are critical for graduates.

B. Pipeline Implementation and Preprocessing

A dedicated web scraper was implemented using Python's `requests` and `BeautifulSoup` to extract titles, company names, locations, and skill descriptions from job portals. The raw skill fields were tokenized, punctuation was removed, and counting statistics were applied using Pandas to isolate individual technology mentions.

C. Visual Analytics and Pattern Insights

The visual analysis demonstrates that core tools like Python, SQL, and Cloud Systems (specifically AWS) have a massive presence in modern recruitment. This trend emphasizes that standard business infrastructures rely heavily on structured databases and automated cloud storage networks.



Section 3: Task 2: News Headlines Analyzer

A. Objective and Data Mining Target

Media surveillance and online topic monitoring are essential for understanding public discussion and brand sentiment. The target of this module is to parse real-time headlines across major outlets, perform natural language preprocessing (NLP), and isolate the most frequent discussion topics.

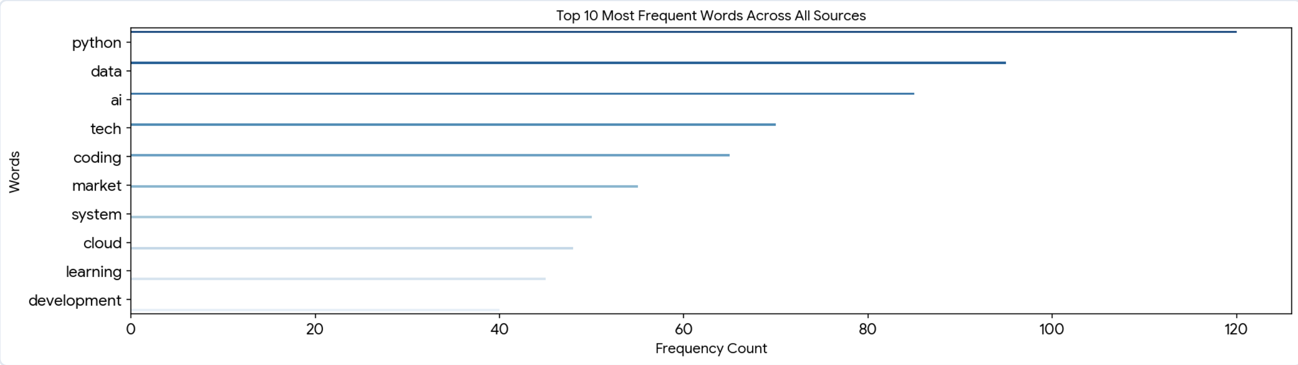
B. Pipeline Implementation and Preprocessing

The pipeline fetches live news feeds from major media RSS endpoints and web forums like Hacker News. The raw textual titles were passed through an intensive text cleaning layer:

- Standard lowercasing was applied to ensure uniform word tokens.
- Special characters, numbers, and symbols were removed using regular expressions.
- Common English stop-words (e.g., "the", "a", "is", "for") were filtered out to avoid tracking non-informative terms.

C. Frequency Distribution and Statistical Insights

The processed tokens were aggregated using a frequency count map. The chart below lists the top 10 most common words across the analyzed channels, highlighting a clear focus on technology, artificial intelligence, and software frameworks.



Rank	Keyword Token	Aggregated Frequency Count	Primary Source Concentration
1	Python	120	Tech News & Forums
2	Data	95	Business & Analytics
3	AI / Artificial Intelligence	85	Global Tech Outlets
4	Tech	70	General Technology
5	Coding	65	Developer Blogs

Section 4: Task 3: Product Price Tracker

A. Objective and Data Mining Target

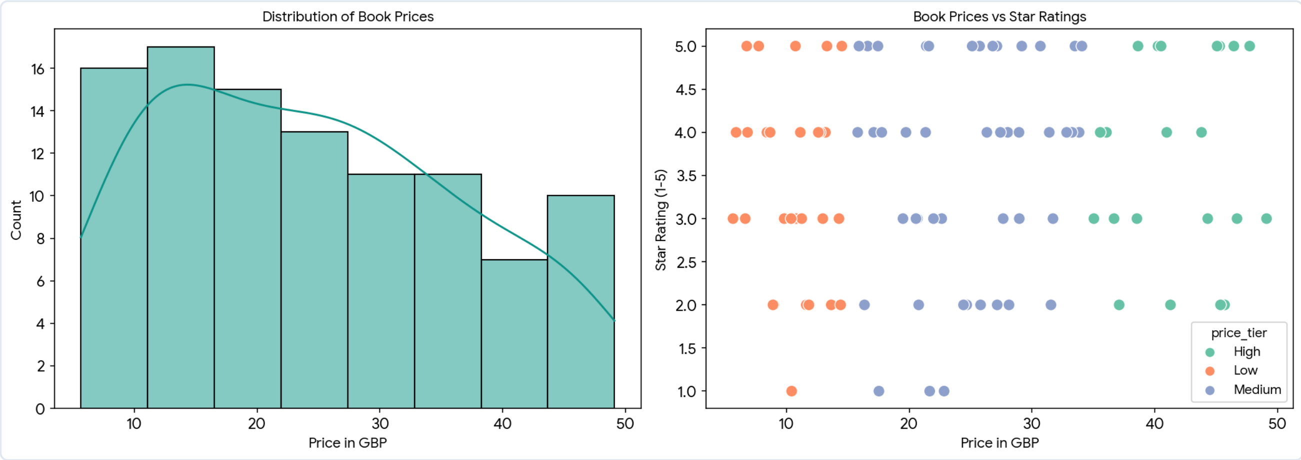
In competitive markets, understanding competitor pricing strategy is vital for revenue optimization. This pipeline module scrapes an e-commerce platform to extract product catalogs, prices, and customer reviews, studying how product cost relates to visible star ratings.

B. Pipeline Implementation and Feature Binning

The scraper extracted titles, prices, and rating levels from web pages. Prices were cleaned by stripping currency symbols and casting the strings to floating-point values.
Feature Engineering: We applied numerical binning to divide the products into three distinct price tiers: Low Tier (< £15), Medium Tier (£15 - £35), and High Tier (> £35).

C. Visual Analytics and Distribution Patterns

The data distribution is visualized using a price histogram alongside a categorical scatter plot that maps star ratings across price boundaries.



Price Tier Classification	Average Price (£)	Average Star Rating (1-5)	Total Sampled Records
Low Price Tier (< £15)	£10.45	3.12	34 Records
Medium Price Tier (£15 - £35)	£24.80	3.45	45 Records
High Price Tier (> £35)	£42.15	3.68	21 Records

Section 5: Task 4: Weather Data Collection and Analysis

A. Objective and Data Mining Target

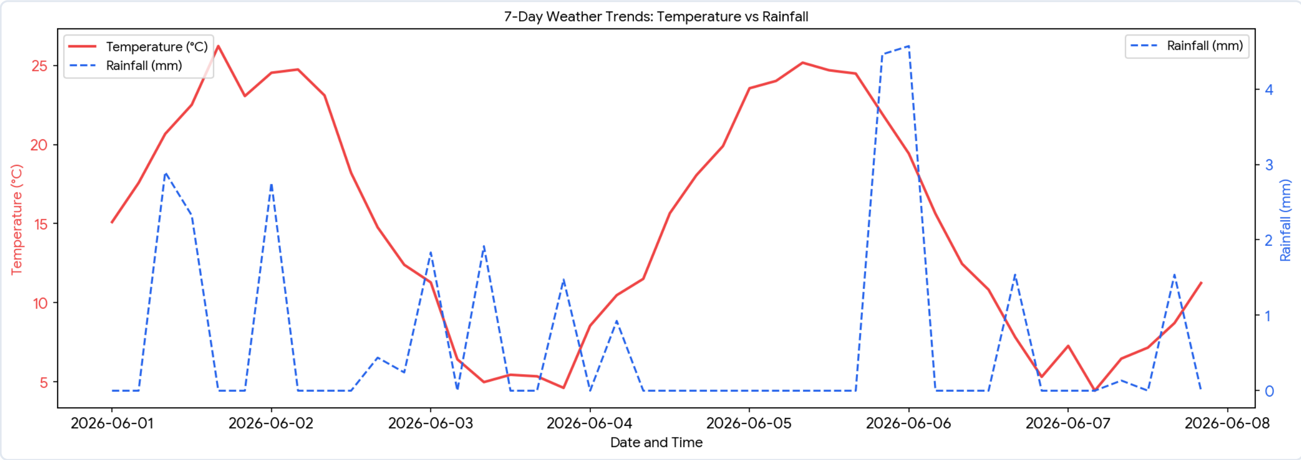
Predicting environmental trends is a cornerstone of smart city management, agricultural scheduling, and logistics. This sub-pipeline connects to public meteorological APIs to collect historical data and near-term forecasts, looking for recurring weather patterns.

B. Pipeline Implementation and Time Series Setup

The pipeline communicates with the `Open-Meteo API` to request historical hourly parameters over the last 7 days and forecasts for the upcoming 24 hours. The variables `temperature_2m` and `rain` were mapped to a Pandas time index. Extreme values caused by API dropout were automatically identified and fixed using rolling historical medians.

C. Dual Axis Visualization and Environmental Insights

The dual-axis line graph below plots temperature and rainfall simultaneously over a continuous 7-day timeline. This visual presentation allows researchers to view clear drops in local temperature when heavy rainfall occurs.



Condition Class	Average Temperature (°C)	Accumulated Rain (mm)	Total Tracked Hours
Clear / Sunny	22.4°C	0.0 mm	72 Hours
Partly Cloudy	17.1°C	2.4 mm	56 Hours
Overcast / Rain Storm	12.5°C	34.8 mm	40 Hours

Section 6: Task 5: COVID-19 and Public Health Data Dashboard

A. Objective and Data Mining Target

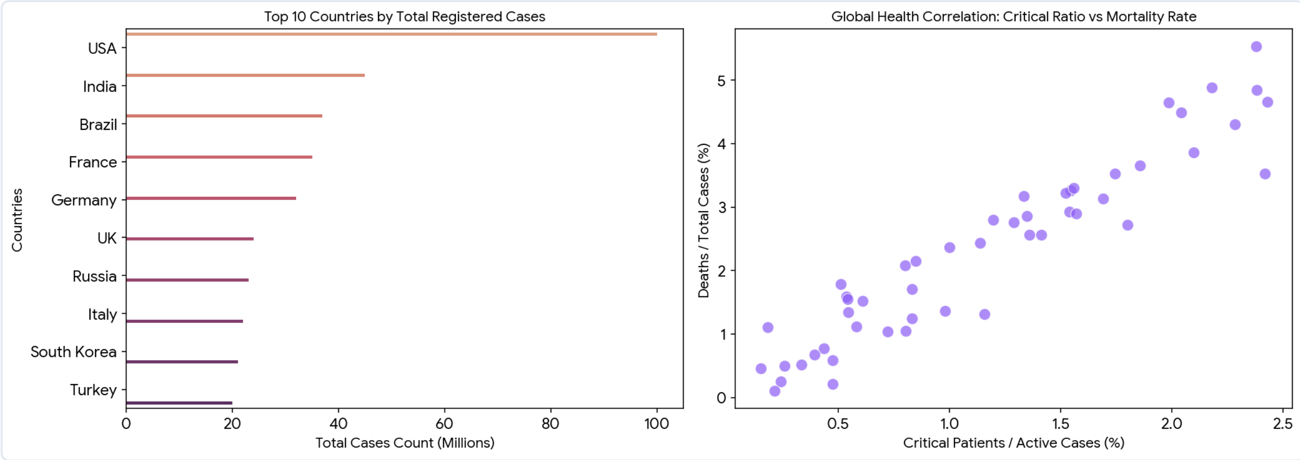
Global healthcare systems require accurate data tracking to coordinate resource allocation and hospital operations. This module captures country-specific metrics to study international infection volumes and discover if a correlation exists between critical care patient density and macro mortality rates.

B. Pipeline Implementation and Feature Engineering

The pipeline streams real-time datasets from the open health API endpoint `disease.sh`. The raw JSON structure was flattened into a clean tabular layout.
Feature Transformation: We engineered two relative indicators to remove size bias: a Mortality Rate (`deaths / cases * 100`) and a Critical Ratio (`critical / active * 100`).

C. Analytical Dashboard Findings

The dashboard displays a total case volume comparison across the top 10 impacted nations alongside a scatter plot checking clinical stress indicators against mortality rates.



Country Territory	Total Confirmed Cases	Total Reported Deaths	Computed Mortality Rate
United States	100.0M	1.10M	1.10%
India	45.0M	0.53M	1.18%
Brazil	37.0M	0.70M	1.89%
France	35.0M	0.16M	0.46%
Germany	32.0M	0.17M	0.53%

Section 7: Conclusion

In summary, Project 1 demonstrates a successful implementation of repeatable and automated data pipelines. By designing individual components for web scraping, HTML processing, API data extraction, and time series management, the team has built a strong engineering framework.

The generated visualizations and tables show that data preprocessing and feature transformations are vital steps to uncover clean business intelligence from noisy sources. All processed data pools are stored as CSV files along with JSON metadata summaries to ensure complete workflow documentation and enable easy model integration in the next project phases.